



CALCULATOR APPLICATION

Procedure to perform standard and advanced mathematical operations, including Calculus, Complex numbers, Plotting, and Matrix features.

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Overview of Features (Detailed Flow)

1. The Calculator Application is a modern web-based tool for performing standard arithmetic and advanced mathematical operations.
2. Users can sign up, log in, and access all calculator functionalities online.
3. The app supports standard arithmetic, scientific, calculus, complex, matrix, and deep learning computations.
4. It includes function plotting, history management, and profile customization.
5. The interface is intuitive and responsive, offering a seamless experience across devices.
6. The system stores your calculation history and allows reuse of results.

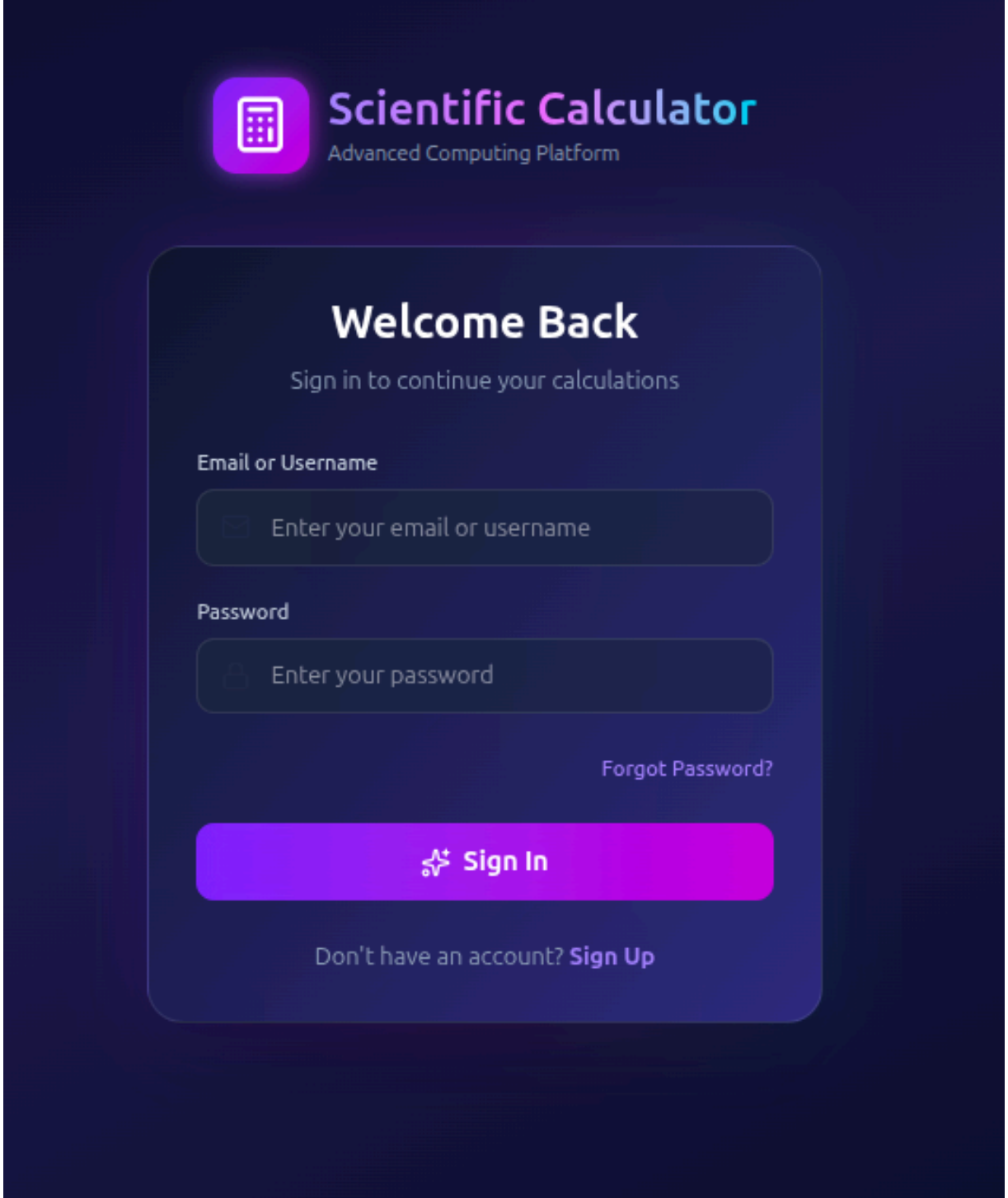
Accessing the Application

- **Signing Up:**

1. Open the application and click Sign Up.
2. Enter your email and click Send OTP.
3. Enter the OTP received in your email to verify.
4. Fill in your full name, username, and password.
5. Click Complete Signup to create your account.

- **Logging In:**

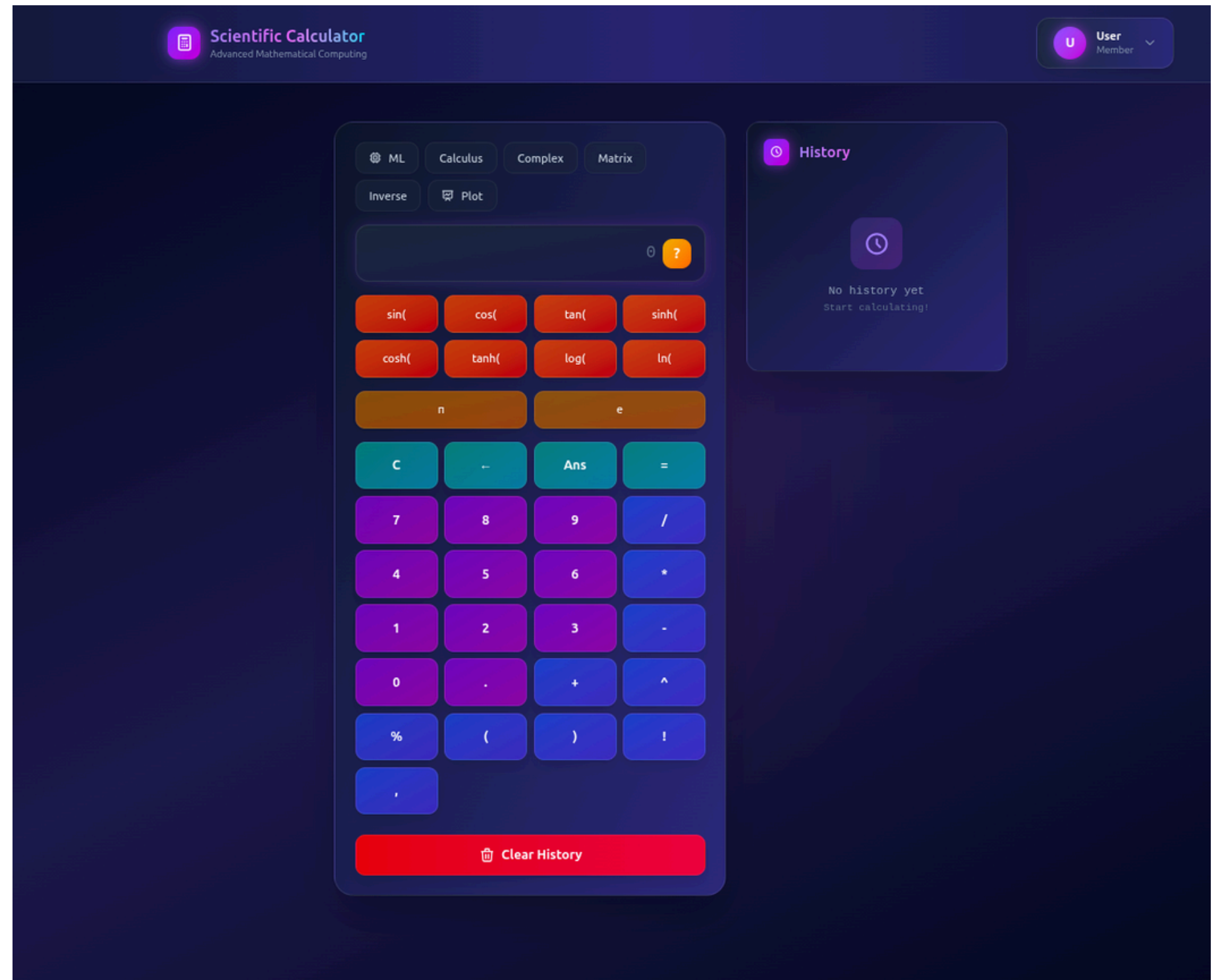
1. Enter your email/username and password.
2. Click Login to access the main calculator.



The screenshot shows the login interface of the 'Scientific Calculator' application. At the top left is a purple calculator icon, followed by the text 'Scientific Calculator' in a light blue font and 'Advanced Computing Platform' in a smaller, lighter blue font below it. The main content area is a dark blue rounded rectangle. Inside, the heading 'Welcome Back' is centered in white, with the subtext 'Sign in to continue your calculations' below it. There are two input fields: 'Email or Username' with a placeholder 'Enter your email or username' and 'Password' with a placeholder 'Enter your password'. A 'Forgot Password?' link is positioned to the right of the password field. A large purple button with a white key icon and the text 'Sign In' is centered below the fields. At the bottom, a link reads 'Don't have an account? Sign Up'.

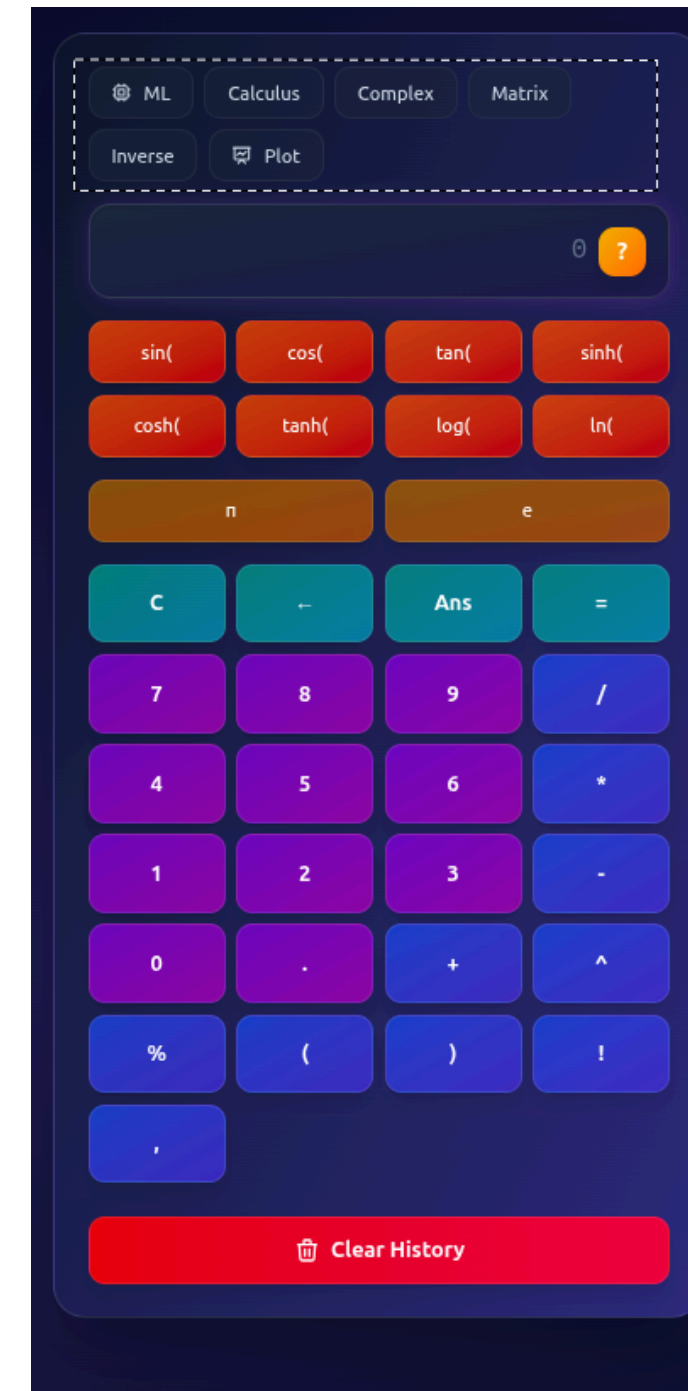
Calculator Interface Overview

- Calculator Input :Main display for entering and viewing expressions/results.
- Side Panel:Displays History or Function Plot Area.
- Profile Dropdown:Allows Sign Out and switching Angle Modes (Radians/Degrees).



Actions Available to the User

- Perform arithmetic operations: +, -, *, /, ^, %, !
- Use constants: π , e
- Access advanced modes:
 1. DL (Deep Learning)
 2. Calculus
 3. Complex
 4. Matrix
 5. Inverse Functions
 6. Plotting
- Manage history, profile, and angle modes.



Main Keypad Buttons

- = → Evaluate expression or complete DL parameter collection
- C → Clear all input or cancel DL sequence
- ← → Backspace last character
- Ans → Insert last result
- 0–9, ., +, -, *, /, ^, %, !, (,), [,], , → Input and structure expressions



Constant and Toolbar Buttons

- Constants:
 1. π (Pi ≈ 3.14159)
 2. e (Euler's number ≈ 2.71828)
- Toolbar Buttons:
 1. DL
 2. Calculus
 3. Complex
 4. Matrix
 5. Inverse
 6. Plot



Deep Learning (DL) Mode

1. Click DL to enable Deep Learning mode.
2. Choose the operation (e.g., C→O, P→O).
3. Enter prompted parameters and press = after each.
4. The calculator displays final results automatically.

DL Mode Functions:

- C→O – Computes the output size of a 1D convolution layer.
- P→O – Computes the output size of a 1D pooling layer.
- PSAME – Calculates the 'SAME' padding needed for 1D convolution or pooling.
- PDOUT – Determines the padding required to achieve a specific 1D output size.
- TC→O – Computes the output size of a 1D transposed convolution layer.
- C1D→P – Calculates the number of parameters in a 1D convolution layer.
- TC1D→P – Calculates the number of parameters in a 1D transposed convolution layer.
- C2D→P – Calculates the number of parameters in a 2D convolution layer.
- TC2D→P – Calculates the number of parameters in a 2D transposed convolution layer.



Calculus Mode

- Toggle Calculus Mode for derivative and integral operations.
- Buttons:
 1. $d/dx()$ → Derivative, e.g., $d/dx(x^3, x)$
 2. $\int()$ → Integral, e.g., $\int(x^2, x, 0, 1)$
 3. x → Shortcut for variable



Complex Mode

- Toggle Complex Mode for imaginary operations.

- Buttons:

1. i

2. re(

3. im(

4. conj(

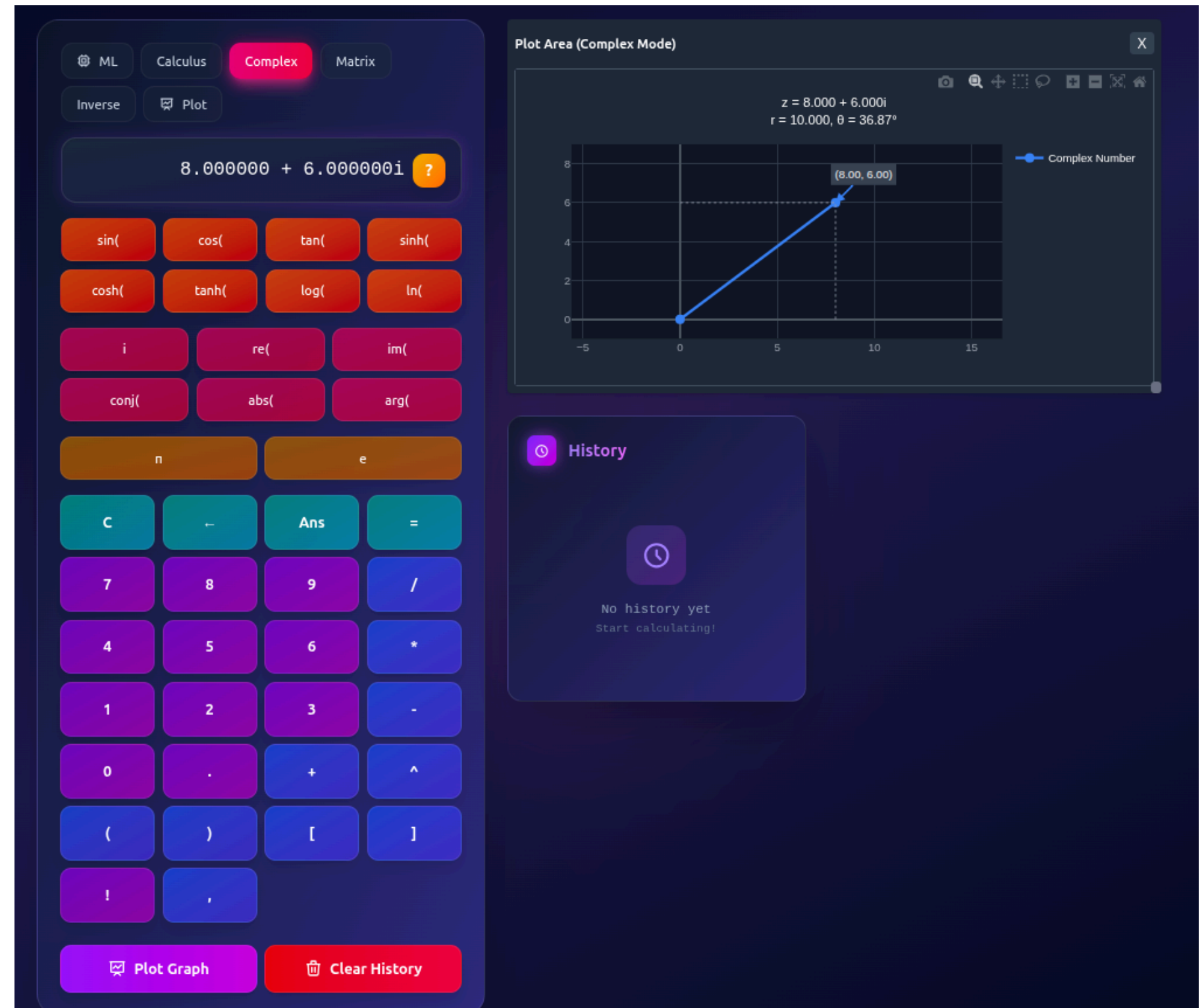
5. abs(

6. arg(

- Examples:

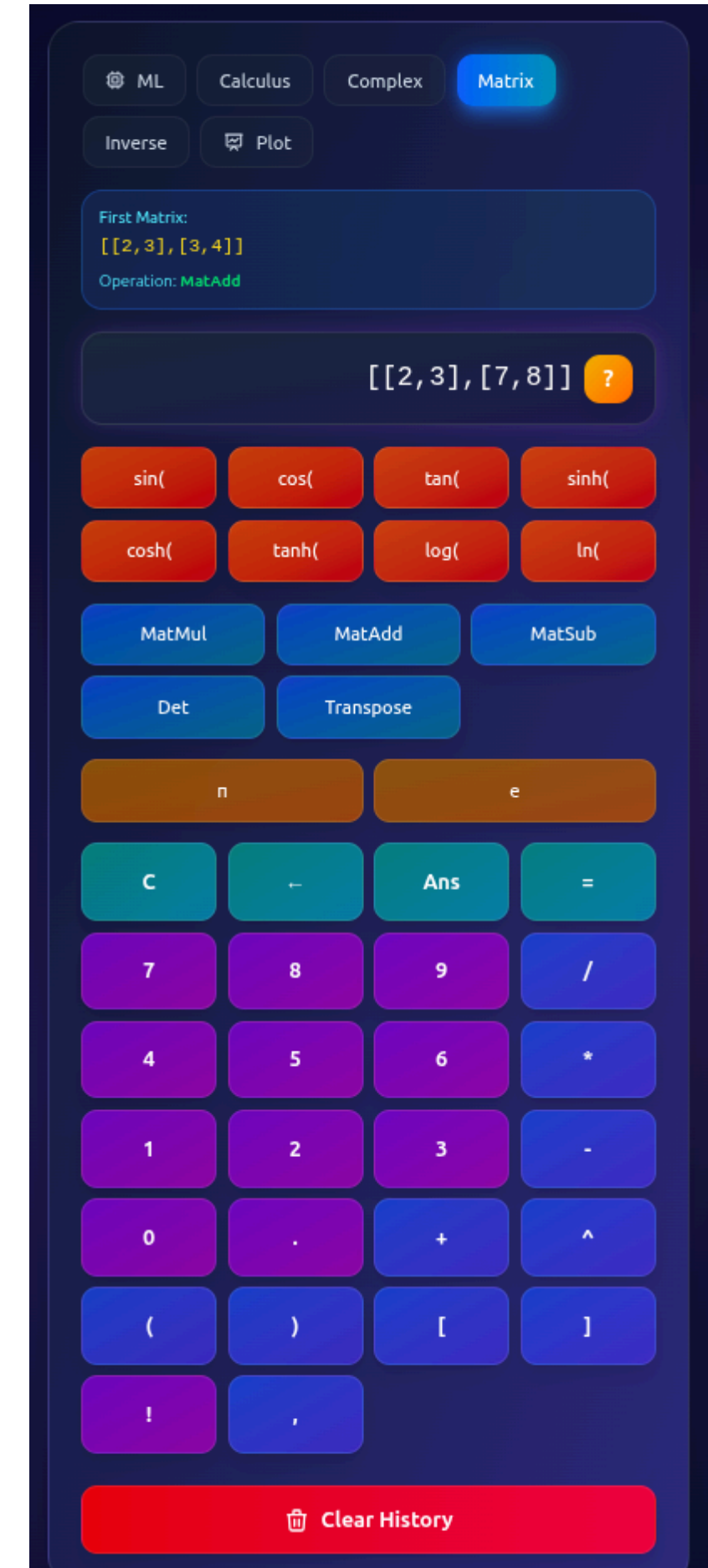
- $\text{re}(5 + 2i) \rightarrow 5$

- $\text{abs}(3 + 4i) \rightarrow 5$



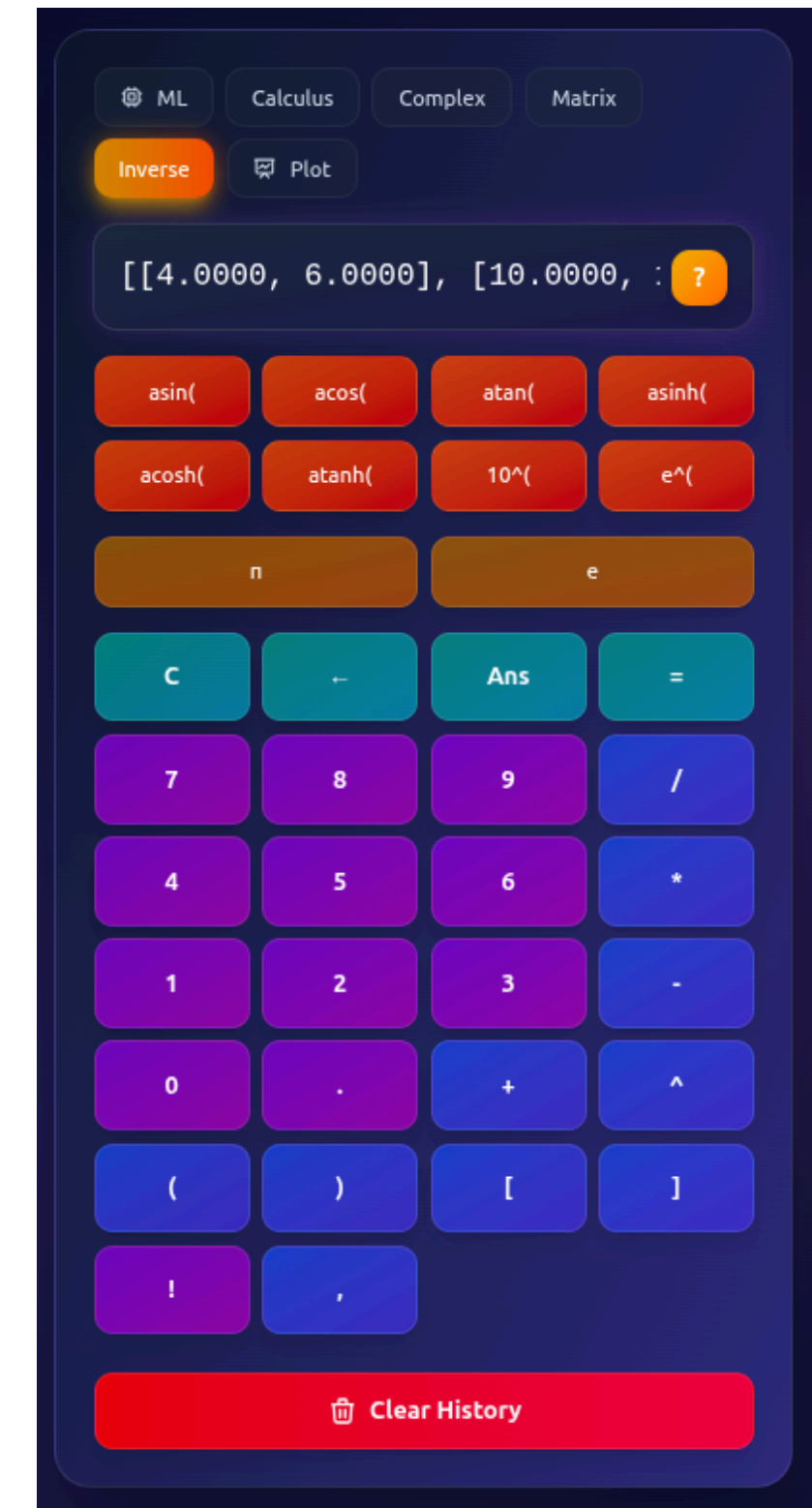
Matrix Mode

- Click Matrix to open the Matrix Modal.
- Define matrices (A, B).
- Choose operations: Add, Multiply, Inverse, etc.
- The result is displayed in the main calculator and saved to history.



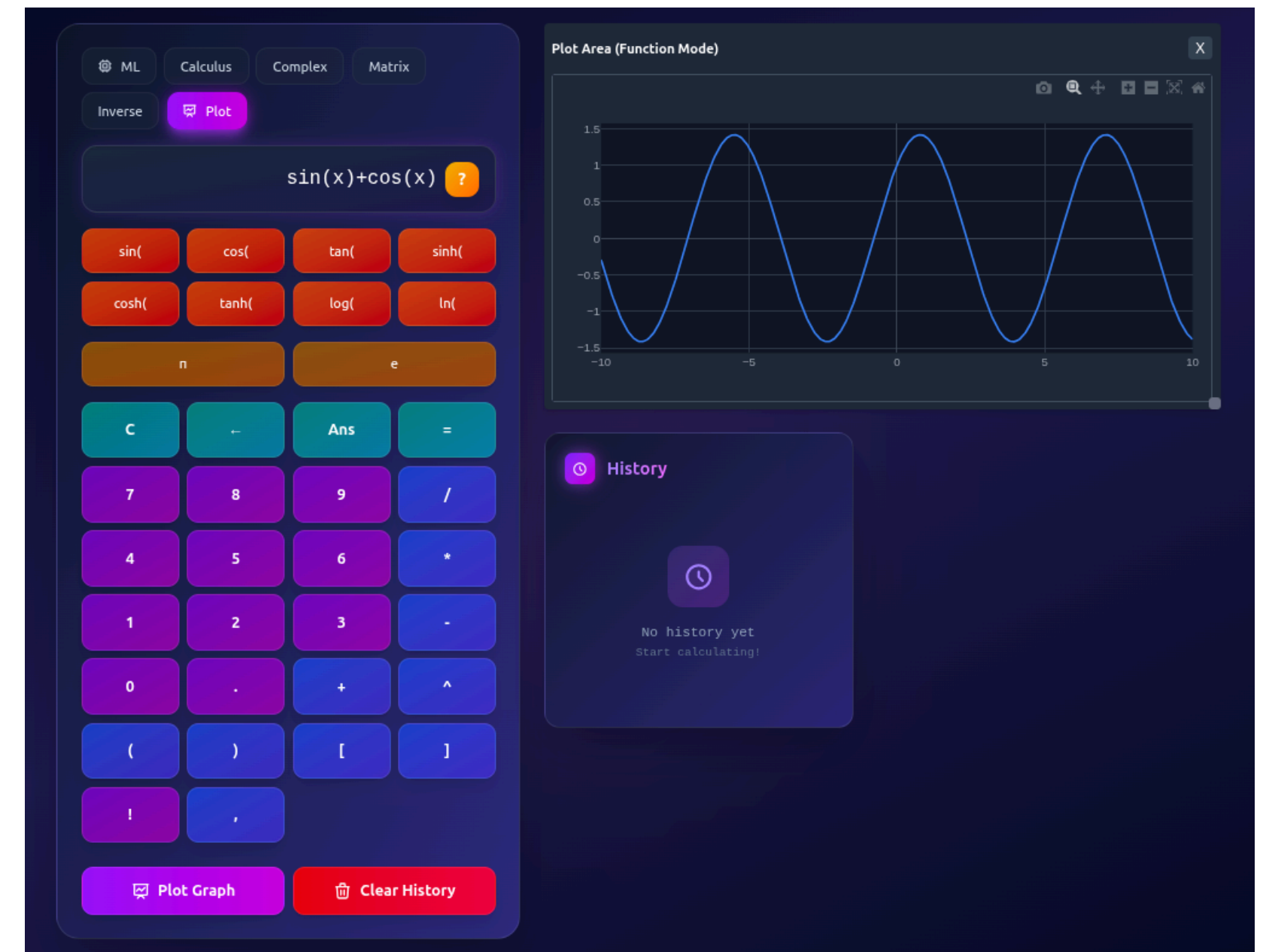
Inverse & Advanced Functions

- Inverse Toggle: Switch between standard and inverse forms.
- Normal Mode:
 $\sin()$, $\cos()$, $\tan()$, $\log()$, $\ln()$, $\sinh()$, $\cosh()$, $\tanh()$
- Inverse Mode:
 $\operatorname{asin}()$, $\operatorname{acos}()$, $\operatorname{atan}()$, $e^{\wedge}()$, $\operatorname{asinh}()$, $\operatorname{acosh}()$, $\operatorname{atanh}()$



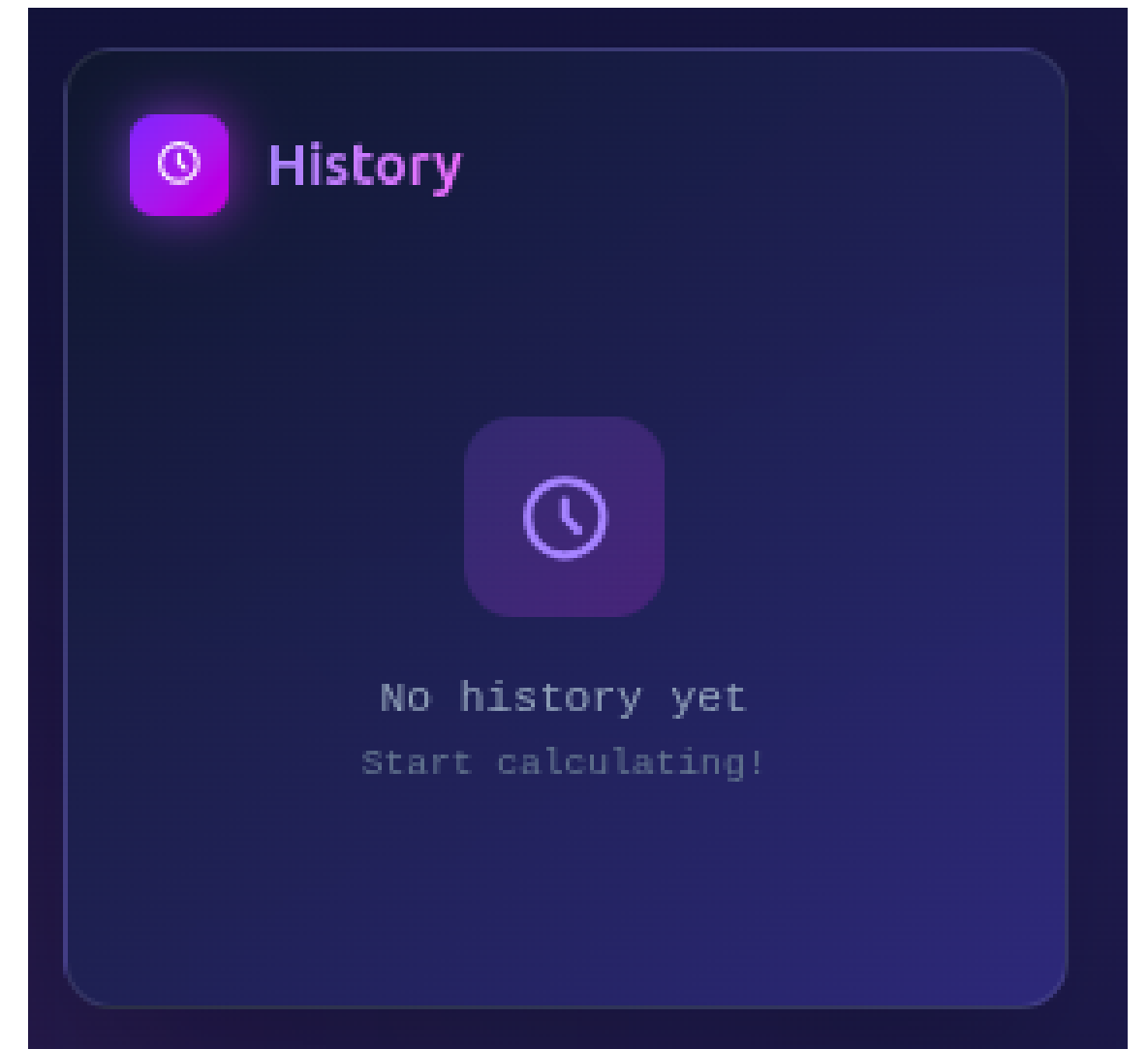
Plot Mode

- Click Plot to open the function graph area.
- Enter a function of x (e.g., $x^2 - 1$).
- The plotted graph appears in the side panel.
- Close the view when done.



History Panel

- Shows list of recent calculations.
- Click a past expression to reuse it.
- Clear History: Removes all stored records and resets Ans.



Profile Dropdown

- Located at top-right corner.
- Options:
 - Mode: Radians / Degrees
 - Sign Out: Logout and return to login screen
- Controls global settings for trigonometric evaluations.